WORKED SOLUTION

STEP II 2003, Question 8

Pure • Differential equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q8

The key idea

The equation is first order and separable: $\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}t}$ equals a rational function of t. Dividing the quadratic by $t+1$ splits that function into a polynomial plus a simple fraction, each of which integrates at once. The stationary points then come straight from the differential equation itself, with no need to differentiate the solution: $\frac{\mathrm{d}y}{\mathrm{d}t}=0$ exactly where the quadratic t^2-3t+2 vanishes.

Solution

Finding y

Polynomial division gives

$$\frac{t^2-3t+2}{t+1}=t-4+\frac{6}{t+1},$$

since $(t+1)(t-4)+6=t^2-3t-4+6=t^2-3t+2$.

The solution must start at $y=A>0$. If y were ever zero, then by uniqueness of solutions to this linear equation it would be the zero solution, which contradicts $y(0)=A$. So $y>0$ throughout (on $t>-1$, where the coefficient is defined), and we may divide by y:

$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}t}=-k\left(t-4+\frac{6}{t+1}\right).$$

Integrating from 0 to t, and using $\ln 1=0$,

$$\ln\frac{y}{A}=-k\left(\frac{t^2}{2}-4t+6\ln(t+1)\right),$$

so

$$\boxed{y=A\,(t+1)^{-6k}\,\mathrm{e}^{-k(t^2/2-4t)}=A\,(t+1)^{-6k}\,\mathrm{e}^{k(8t-t^2)/2}}.$$

The stationary values

From the differential equation,

$$\frac{\mathrm{d}y}{\mathrm{d}t} = -k\,\frac{(t-1)(t-2)}{t+1}\,y.$$

We assume $k \neq 0$ (otherwise $y \equiv A$ and every point is stationary). Since $y > 0$ and $t+1 > 0$, the derivative vanishes only at $t = 1$ and $t = 2$, and it changes sign at each of them because each is a simple root. So there are exactly two stationary values:

$$y(1) = A \cdot 2^{-6k}\,\mathrm{e}^{-k(1/2-4)} = A \cdot 2^{-6k}\,\mathrm{e}^{7k/2},$$
$$y(2) = A \cdot 3^{-6k}\,\mathrm{e}^{-k(2-8)} = A \cdot 3^{-6k}\,\mathrm{e}^{6k}.$$

Their ratio is

$$\frac{y(1)}{y(2)} = \frac{3^{6k}}{2^{6k}}\,\mathrm{e}^{7k/2-6k} = \left(\frac{3}{2}\right)^{6k}\mathrm{e}^{-5k/2},$$

as required.

Ask yourself Which of the two stationary values is the larger?

Take logarithms:

$$\ln\frac{y(1)}{y(2)} = k\left(6\ln\tfrac{3}{2} - \tfrac{5}{2}\right),$$

and $6\ln 1.5 \approx 2.433 < 2.5$, so the bracket is negative (about -0.067). Hence $y(1) < y(2)$ when $k > 0$ and $y(1) > y(2)$ when $k < 0$. The difference is small, so the wiggle in the graph is gentle. Similarly

$$\ln\frac{y(1)}{A} = k\left(\tfrac{7}{2} - 6\ln 2\right) \approx -0.659k, \qquad \ln\frac{y(2)}{A} = k(6 - 6\ln 3) \approx -0.592k,$$

so both stationary values lie below A when $k > 0$ and above A when $k < 0$.

Behaviour as $t \to +\infty$

Write

$$\ln\frac{y}{A} = -k\left(\frac{t^2}{2} - 4t + 6\ln(t+1)\right).$$

As $t \to \infty$ the bracket tends to $+\infty$, dominated by $\frac{1}{2}t^2$ (the ln term grows far more slowly). So:

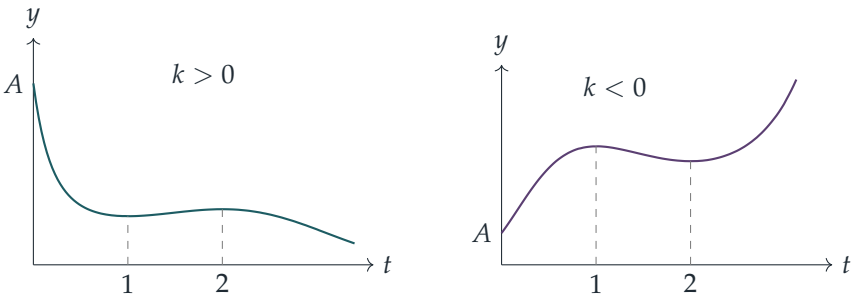
- if $k > 0$, then $\ln(y/A) \to -\infty$, so $y \to 0$ (very rapidly, like $\mathrm{e}^{-kt^2/2}$);
- if $k < 0$, then $\ln(y/A) \to +\infty$, so $y \to \infty$ (like $\mathrm{e}^{|k|t^2/2}$).

Sketches

The sign of $\frac{\mathrm{d}y}{\mathrm{d}t}$ is the sign of $-k(t-1)(t-2)$ for $t > 0$.

- $k > 0$: y decreases from A on $(0,1)$ to a local minimum at $t = 1$, rises to a local maximum at $t = 2$ (still below A), then decreases towards 0.
- $k < 0$: y increases from A to a local maximum at $t = 1$, falls to a local minimum at $t = 2$ (still above A), then increases without bound.

The graphs below are drawn accurately with $A = 1$ and $k = \pm 2$.



Checks

- At $t = 0$ the formula gives $A \cdot 1 \cdot \mathrm{e}^0 = A$. Differentiating $\ln y = \ln A - 6k\ln(t+1) - k(t^2/2 - 4t)$ gives $\frac{y'}{y} = -\frac{6k}{t+1} - k(t-4)$, which recombines to the original equation.
- Numerically, integrating the differential equation step by step (Runge–Kutta) with $A = 1.5$ and $k = \pm 2$ reproduces the formula at $t = 3$ to eight significant figures. With $k = 2$ the ratio $y(1)/y(2)$ is 0.8742, matching $(1.5)^{12}\mathrm{e}^{-5}$.

Common slips

- Getting the division wrong: the remainder is 6, not 2. Multiply back by $t + 1$ to check.
- Inverting the ratio: the target is $y(1)/y(2)$, and the signs in the exponent make it easy to get $y(2)/y(1)$ by mistake.
- Sketching both turning points on the wrong side of A: the earlier calculation places both below A for $k > 0$ and both above A for $k < 0$.

Beyond STEP

The equation is linear, $y' + p(t)y = 0$, and its solution $A\exp\big(-\int_0^t p\big)$ is the simplest case of the integrating-factor method. It also illustrates a general principle you will meet in a first ODE course: you can often read off qualitative features (stationary points, sign of the solution, growth or decay) directly from the equation without solving it. Boyce and DiPrima, *Elementary Differential Equations*, develops both ideas, including the uniqueness theorem we used to show that y never vanishes.

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